

# Cart Switch, Single Bank

The soft-start high-side switch and cartbeat input from `d_interface_cart.h`, on **one bank** of a 5×30 solderable breadboard, with the node's 3V3 regulator aboard. Twelve consecutive rows, no cross-channel jumper, R5 at 10k.

12 rows · one bank

3V3 regulator on board

2 links

9 harness wires

R5 = 10k

no sleeved leads

## 01 What changed, and why the rows land where they do

**R5 is 10k, not 1k.** It is a series resistor into a GPIO and the top leg of a divider against R6's 100k, so the only requirement is that it be much smaller than R6. At 10k the input reads 3.00V against a threshold of 2.47V — half a volt of margin — and the fault current, if the cart's beat pin ever presented 5V, drops from 1mA to 100µA. Anything from 220Ω to 22k works; 47k and above fails.

**The 3V3 regulator is on the board**, because the ESP32's 5V pin is disconnected for safety and the node has to be fed 3V3 instead. It is the same module the battery sheet documents, and it goes in **rows 1, 0 and -1 of column D**: Vi at **1D**, GND at **0D**, Vo at **-1D**. Two rows are added above row 1 and **every existing row keeps its number**, which is the whole reason for counting upward from zero.

**Vi needs no wire at all** — 1D is in the 5V row, so the module taps the rail by being plugged into it. That pushes the incoming 5V lead out to **1E**, the only other free hole in that row. Its ground is the part that costs something: it cannot share a row with the main ground, so row 0 is its own ground row and **L2** links it down to the harness ground at **8A** — which in turn is why the ESP32's own ground wire moves up to **0E**, beside its 3V3 at -1E.

### The same module runs four times hotter here

On the battery sheet it regulates 3.7V straight off the cell. Here it regulates the 5V the solar/battery unit presents, so it drops 1.7V instead of 0.4V. If the module is a linear regulator, dissipation at a 500mA LoRa transmit peak is 850mW against 200mW on the other board, and 136mW against 32mW at idle. A buck converter does not care. Worth knowing which one is in your hand before this goes in a sealed box — and worth checking its current rating covers the transmit peak, not just the idle draw.

**Everything is on one bank now**, and the row order is not a free choice. Four parts have to stand upright across a single row span — R4 R3 R6 R5 — and each

transistor needs three consecutive rows. Those constraints chain into exactly the order below, with one consequence worth knowing: **R2 can never stand upright**. Q2's base is always flanked by its own collector and emitter, so the row carrying CART\_PWR is always two steps away and R2 always spans 0.2".

**Ground is three rows and two links.** Row 6 carries the components that reference it (Q2's emitter, R3, C3), row 8 carries the harness and R6, and row 0 carries the regulator and the ESP32. **L1** joins 6 to 8 at column E; **L2** joins 0 to 8 at column A. Spreading ground over three rows is also what retires the old cross-channel jumper — the extra rows do the job the right bank used to.

### The two links are not the same kind of wire

**L1 (6E→8E) carries almost nothing** — Q2's emitter current and C3's return, microamps and a decoupling transient. A resistor lead offcut is plenty. The cart's heavy return never touches it: the pack ground arrives at **8D** and the cart's ground leaves at **8C**, one hole apart on the same row.

**L2 (0A→8A) carries the node's whole supply return**, LoRa transmit bursts included — so it is harness-gauge insulated wire, run on the **track side**, not an offcut. It has to be insulated whichever way you route it: every hole it passes at column A holds a transistor leg.

## 02 Reinforce three rows before anything else

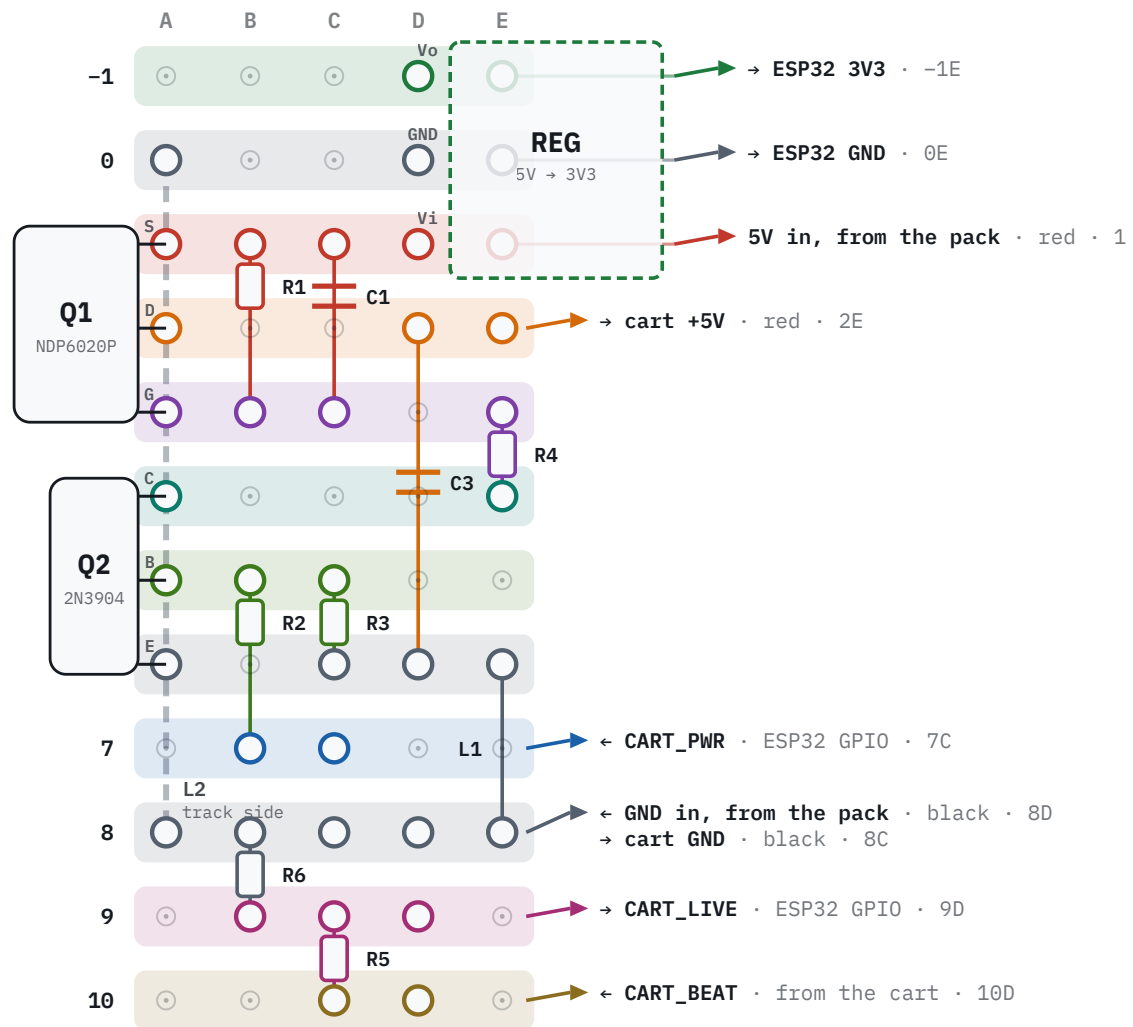
This board carries the cart — a Pi Zero and an LTE modem, whose transmit bursts pull amps — and proto-board strips are thin, narrow copper never intended for it.

**Lay solid tinned wire along rows 0, 1, 2 and 8 on the track side** and solder it to every pad in those rows. Rows 1, 2 and 8 are the cart's path. **Row 0 is there for the node**: its whole supply return crosses that strip from 0E to L2 at 0A. Row 6 is signal-level ground only and needs nothing, and row -1 carries the same current as row 0 but across a single hole's worth of copper. Bring the harness in and out on 20–22AWG, not hookup wire from the signal reel.

Everything else is signal: the gate network runs at tens of microamps and the cartbeat divider at microamps.

## 03 Layout

|             |                    |             |           |              |         |
|-------------|--------------------|-------------|-----------|--------------|---------|
| 3V3 out     | 5V always-on · red | SW_5V · red | GATE      | Q2 collector | Q2 base |
| GND · black | CART_PWR           | CART_LIVE   | CART_BEAT |              |         |



Rows -1 to 10 of one bank, component side. Both transistors sit in column A with their legs in three consecutive rows, and the regulator sits in column D with its three pins in rows 1, 0 and -1. Holes drawn with a dot are deliberately empty: R1, C1, R2 and C3 each fly over one or more of them, and because nothing is in those holes none of those leads needs sleeving. L2 is the exception and is drawn dashed — it runs underneath, on the track side. The regulator's dashed outline is its body, not its holes: it overhangs column E and the centre channel, where the three wires beneath it are.

| ROW | NODE            | HOLES                                                                        |
|-----|-----------------|------------------------------------------------------------------------------|
| -1  | 3V3 out         | -1D REG Vo · -1E to the ESP32's 3V3 pin                                      |
| 0   | GND — regulator | 0A L2 down to 8A · 0D REG GND · 0E to ESP32 GND                              |
| 1   | 5V always-on    | 1A Q1 source · 1B R1 · 1C C1 · 1D REG Vi · 1E 5V in from the pack            |
| 2   | SW_5V           | 2A Q1 drain · 2D C3 · 2E to cart +5V — 2B, 2C stay empty: R1 and C1 fly over |
| 3   | GATE            | 3A Q1 gate · 3B R1 · 3C C1 · 3E R4 — 3D empty, C3 flies over                 |
| 4   | Q2 collector    | 4A Q2 collector · 4E R4 — 4D empty                                           |

|    |               |                                                                                                                     |
|----|---------------|---------------------------------------------------------------------------------------------------------------------|
| 5  | Q2 base       | <b>5A</b> Q2 base · <b>5B</b> R2 · <b>5C</b> R3 — <i>5D empty</i>                                                   |
| 6  | GND — signal  | <b>6A</b> Q2 emitter · <b>6C</b> R3 · <b>6D</b> C3 · <b>6E</b> L1 to 8E — <i>6B empty, R2 flies over</i>            |
| 7  | CART_PWR      | <b>7B</b> R2 · <b>7C</b> from the node's CART_PWR pin — <i>7E empty, L1 flies over</i>                              |
| 8  | GND — harness | <b>8A</b> L2 from 0A · <b>8B</b> R6 · <b>8C</b> to cart GND · <b>8D</b> GND in from the pack · <b>8E</b> L1 from 6E |
| 9  | CART_LIVE     | <b>9B</b> R6 · <b>9C</b> R5 · <b>9D</b> to the node's CART_LIVE pin                                                 |
| 10 | CART_BEAT     | <b>10C</b> R5 · <b>10D</b> from the cart's cartbeat pin                                                             |

## 04 Parts

| REF        | VALUE                  | HOLES         | SPAN | NOTES                                                        |
|------------|------------------------|---------------|------|--------------------------------------------------------------|
| <b>Q1</b>  | NDP6020P P-FET, T0-220 | 1A / 2A / 3A  | —    | legs into rows 1, 2, 3 = <b>S, D, G</b> — see section 6      |
| <b>Q2</b>  | 2N3904 NPN, T0-92      | 4A / 5A / 6A  | —    | legs into rows 4, 5, 6 = <b>C, B, E</b>                      |
| <b>R1</b>  | 100k                   | 1B → 3B       | 0.2" | flat — holds the gate at the source                          |
| <b>C1</b>  | 1u                     | 1C → 3C       | 0.2" | flat film is a perfect fit; splay a ceramic                  |
| <b>R4</b>  | 10k                    | 3E → 4E       | 0.1" | stand upright                                                |
| <b>R3</b>  | 100k                   | 5C → 6C       | 0.1" | stand upright                                                |
| <b>R2</b>  | 10k                    | 5B → 7B       | 0.2" | flat — cannot be upright, see section 1                      |
| <b>C3</b>  | 100n                   | 2D → 6D       | 0.4" | splay the legs — local decoupling at the FET                 |
| <b>L1</b>  | wire offcut            | 6E → 8E       | 0.2" | joins signal ground to harness ground; microamps             |
| <b>R6</b>  | 100k                   | 8B → 9B       | 0.1" | upright now — the load-bearing pull-down                     |
| <b>R5</b>  | 10k                    | 9C → 10C      | 0.1" | stand upright — <b>was 1k</b>                                |
| <b>REG</b> | 3V3 regulator module   | 1D / 0D / -1D | 0.2" | pins into rows 1, 0, -1 = <b>Vi, GND, Vo</b> — see section 6 |
| <b>L2</b>  | insulated, 20–22AWG    | 0A → 8A       | 0.8" | <b>track side</b> — carries the node's whole supply return   |
| <b>C2</b>  | 100u                   | —             | —    | <b>not on this board</b> — fit it at the cart                |

**C2 is deliberately absent.** The modem's peaks are faster than anything upstream can answer, so the bulk belongs where the current is drawn. A 100µF at the cart end of the harness does its job; the same part here, diverging from the FET and a

end of the harness does its job, the same part here, six rows from the FET and a harness away from the load, mostly does not. C3 stays because local decoupling belongs at the FET.

## 05 Solder order

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The power rows get reinforced first, while nothing is in the way of the iron.

- 1** Cut the board to 14 rows, keeping **one** bank, and mark row 1 — you are numbering two rows *above* it. Lay and solder the reinforcing wire along rows 0, 1, 2 and 8, then fit **L2** on the track side from 0A to 8A while the underside is still clear.  
Check — **1↔2 open, 1↔8 open, 0↔8 near zero**, and each reinforced row continuous end to end.
- 2** Fit **Q2** into 4A/5A/6A, **R4** upright (3E–4E) and **R3** upright (5C–6C).  
Check — **diode-test Q2, 3↔4 = 10k, 5↔6 ≈ 100k**.
- 3** Fit **R1** (1B→3B), **C1** (1C→3C), **R2** (5B→7B) and **C3** (2D→6D). Keep 2B, 2C, 3D, 4D, 5D and 6B empty — those are the lanes.  
Check — **1↔3 settles at 100k, 5↔7 = 10k, 2↔6 climbs to open**.
- 4** Fit **Q1** with its leads bent so the body lies off the board edge.  
Check — the **body-diode orientation test** from section 6, and **1↔6 still open**.
- 5** Fit **L1** (6E→8E), then **R6** upright (8B→9B) and **R5** upright (9C→10C).  
Check — **6↔8 near zero, 8↔9 = 100k, 9↔10 = 10k**.
- 6** Land the nine harness wires: 1E red in from the pack, 2E red out to the cart, 8C and 8D black, –1E and 0E to the node's 3V3 and GND, 7C and 9D to its GPIOs, 10D from the cart's beat pin. Do –1E, 0E and 1E **before** the module goes on, and dress them clear — its body sits over them.  
Check — **1↔8 open** one last time, with nothing plugged in, before the pack goes anywhere near it.
- 7** Fit the **regulator** last, pins into 1D / 0D / –1D, body toward column E. **Then power the pack and measure row –1 against row 0 before the node is connected to anything.**  
Check — **3.3V ±0.1**, and **–1↔1 not near zero**. See the warning in section 7.

## 06 Mounting Q1 and the regulator

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A TO-220 is a big part on a 0.1" grid. Its leads are on the right pitch and drop straight into rows 1–3, but the body standing upright is 10mm across — four rows' worth — so it would sit over Q2. **Bend the leads 90° just below the body so it lies off the top edge of the board, tab outward.** Nothing is then underneath it.

### Two things about this part specifically

**The tab is the DRAIN, not ground.** On this P-FET the tab sits at SW\_5V, so it must not touch the chassis, the USB-C shell, or the board. If the chassis is metal, insulate it or point it into free air.

**Swapping source and drain leaves the cart permanently on,** and the gate drive appears to do nothing — the body diode conducts drain-to-source and simply passes 5V around the channel. It is a genuinely confusing failure, and one meter reading rules it out. In diode mode, red on row 2, black on row 1 should read about 0.5V, and the reverse should be open. If it reads the other way round, the part is in backwards.

## Mounting the regulator

Its pins go into **1D, 0D and -1D** and its body points toward column E, out over the centre channel, where the only things under it are the three harness wires at -1E, 0E and 1E. **Stand it off the board by two or three millimetres** rather than pulling it flush — that clears those wires, and keeps the module's underside off the unused right bank's copper if the body reaches that far. A strip of tape on the right bank costs nothing and settles it.

Pointing the body the other way, over columns A to C, is the obvious-looking choice and the wrong one: at row 1 it would sit directly on Q1's bent lead and on R1's and C1's.

Heat is not a concern for Q1: a few hundred milliwatts at the current the cart draws, in a package rated for tens of watts. No heatsink, and the tab can stay bare as long as it touches nothing. The one specification that matters is the one the header names — logic level, with  $R_{DS(on)}$  quoted at  $V_{GS} = -4.5V$ , which the NDP6020P is.

## 07 Bench tests

### Resistance, unpowered

| ROWS  | EXPECT | NOTES                                                             |
|-------|--------|-------------------------------------------------------------------|
| 1 ↔ 3 | 100k   | R1, climbing as C1 charges — start high-range and watch it settle |
| 3 ↔ 4 | 10k    | R4                                                                |

|        |               |                                                                                                                                                                                                                                        |
|--------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 ↔ 6  | ≈100k         | R3, in parallel with Q2's base-emitter junction                                                                                                                                                                                        |
| 5 ↔ 7  | 10k           | R2                                                                                                                                                                                                                                     |
| 2 ↔ 6  | open          | C3 only — a cap, so the reading climbs away to open                                                                                                                                                                                    |
| 6 ↔ 8  | ≈0Ω           | L1 — meant to be a dead short                                                                                                                                                                                                          |
| 0 ↔ 8  | ≈0Ω           | L2 — likewise. If this one is open the node has no ground return and nothing works                                                                                                                                                     |
| 8 ↔ 9  | 100k          | R6                                                                                                                                                                                                                                     |
| 9 ↔ 10 | 10k           | R5 — <b>10k now, not 1k</b>                                                                                                                                                                                                            |
| -1 ↔ 1 | not near zero | <b>3V3 rail to the 5V rail, through the module.</b> The exact reading depends on the module and means little; near zero means 5V reaches the node's 3V3 pin and kills it. Measure with the module fitted and before the node is wired. |
| 1 ↔ 8  | open, >1M     | <b>5V to harness ground.</b> The one fault that can cook the pack. Check <b>1↔6</b> and <b>1↔0</b> as well — with both links in, all three ground rows are one net.                                                                    |
| 2 ↔ 1  | a diode       | Q1's body diode — see the orientation test in section 6                                                                                                                                                                                |

### Measure 3V3 before the node is connected to it

The module's pins are Vi–GND–Vo in that order, and once it is soldered its body gives no clue which row is which. Fitted the wrong way up, or if it is not the part you think it is, row -1 sits at 5V — straight onto the node's 3V3 pin, which does not survive it. Power the pack with the node unplugged, put the meter on **row -1 against row 0**, and look for 3.3V. Anything above 3.6V means stop.

### Powered, no cart attached

Bench supply at 5.0V with the limit at 100mA, and drive CART\_PWR from a 3.3V source rather than the node. A dummy load — a 47Ω resistor on SW\_5V, about 100mA — is worth fitting so the ramp has something to work against.

| STATE                    | EXPECT                         | NOTES                                                                                                                                                            |
|--------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pack on, node unplugged  | row -1 = 3.3V                  | the regulator, before anything depends on it. Repeat it with the node drawing current: a module that sags under a LoRa transmit burst is undersized, not faulty. |
| CART_PWR low or floating | SW_5V = 0V                     | Q2 off, R1 holds the gate at the source. Draw from 5V is leakage only — unmeasurable on a handheld meter.                                                        |
| CART_PWR at 3.3V         | SW_5V reaches 5V in about 10ms | R4 and C1 set the ramp. ~50μA from 5V through R4, plus ~260μA sourced by the GPIO through R3                                                                     |

|                      |                          |                                                                                                                               |
|----------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 3.3V                 |                          | R1, plus ~200µA sourced by the GPIO through R2.                                                                               |
| CART_PWR<br>back low | SW_5V decays over ~100ms | R1 alone recharging C1, plus whatever the load holds. Nothing waits on it.                                                    |
| pull 10D to<br>3.3V  | CART_LIVE ≈ 3.00V        | the 10k and 100k divider. <b>Was 3.27V with a 1k.</b><br>Release it and the node should see a clean zero rather than a float. |

If you have a scope, the 10ms ramp on SW\_5V is the one thing worth looking at, because it is the whole reason this is a soft-start switch and not just a transistor.

## 08 The USB data lines

The Pi is the USB host and the ESP32 the device, so the ESP32 asserts a 1.5k pull-up from D+ to its own 3.3V rail. With the ESP32 permanently powered and the cart off, that pull-up pushes current through D+ into the Pi's USB pins, through their ESD clamps and into the Pi's 3.3V rail — roughly  $(3.3 - 0.7) / 1.5k \approx 1.7mA$ . That will not boot a Pi Zero, but it is more than enough to hold a PMIC somewhere indeterminate, which is exactly the “it will not stay off” symptom.

### Cut VBUS, but keep the cable ground

Cutting VBUS between the two boards is essential, though the reason has moved. It used to be that the ESP32's permanent 5V would feed the cart directly and this whole board would do nothing. With the ESP32's 5V pin disconnected that path is gone and the leak runs the other way: VBUS would push the cart's 5V back into the ESP32's 5V net and its onboard regulator's input, while the 3V3 rail is already being driven externally. Cut it either way. But cutting the cable's GND buys you nothing: ground is already common through the power harness, so there is no isolation to gain, and the data pair loses the return path that runs beside it. Keep it.

### If it does leak, try software before the 4066

The node is the thing switching the cart, so it already knows when the cart is off — and when the cart is off, nothing is listening on that USB. So have it release the D+ pull-up for exactly that period. On the C3 the USB Serial/JTAG PHY has a pull-up override (the `usb_serial_jtag_ll_phy_*` calls, over `USB_SERIAL_JTAG_CONF0_REG`); uninstalling the driver gets you to the same place. That removes the only leak path there is, costs no parts, no board space and no signal integrity, and it hangs naturally off `cart_open()` and `cart_close()`, which already bracket the window.

If it has to be hardware, **the 4066 is the wrong part for this one.** A CD4066 is 100–300Ω on-resistance with tens of picofarads of channel capacitance; USB full-speed wants something near 45Ω and a few pF. A 74HC4066 is better, around 30–50Ω at

4.5V, and might well pass 12Mbps — but it is outside what the part is characterised for, which makes it precisely the wrong kind of component to have between you and a site you cannot drive to. A purpose-made USB switch such as a TS3USB221A is a few ohms and a few pF, costs about the same, and is specified for the job.

**Measure before building either.** Power the ESP32, leave the cart switched off, and put a meter in series with the Pi's 5V or 3.3V rail. If you read tens of microamps, the problem does not exist and you can stop. If you read a milliamp or two, the software fix almost certainly ends it.

## 09 Notes

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- **R5 is not a precision part.** It limits current into the GPIO and forms the top leg of a divider against R6; the only constraint is  $R5 \ll R6$ . 10k was chosen because it is in the box and gives better fault protection than 1k, not because the value matters. Two resistors in parallel to synthesise 1k would be two joints and a taller part for nothing.
- **The two ground rows are one net.** Row 6 is the components, row 8 is the harness, and the link makes them the same node. When a test says “ground”, either row will do — but probe row 8 for anything carrying current, because that is the row with the copper.
- **Q2 is a 2N3904, and the header now says so too** — the 2N2222A it used to name is gone from both `d_interface_cart.h` and `d_interface_batt.h`. In TO-92 the 2222 family is pinout-split and the metal-can original is reversed again, where the 3904 is unambiguously E–B–C. The divider's pair is a 3904 and a 3906 for the same reason: one footprint to get right instead of two.
- **The ASCII schematic reads ambiguously around R4.** Drawn as it stands, R4's tee lands on C1's 5V side rather than on the gate, which would leave nothing to pull the gate down. The prose is unambiguous and the two time constants confirm it —  $R1 \times C1 = 100\text{ms}$  for turn-off,  $R4 \times C1 = 10\text{ms}$  for turn-on — so the gate network is R1 and C1 both across 5V to gate, with R4 from gate to Q2's collector, which is what this layout builds. Worth redrawing that corner before it costs someone a board.
- **R6 and R3 are the load-bearing ones.** The header says so and the layout respects it: both hold their nodes in the safe state through reset, flashing, brownout and the window before `cart_init()` runs. If you are ever tempted to drop a part to save a row, not these.
- **The cartbeat and the USB data both reference the harness ground.** That makes the GND splice at the pack's connector a signal path, not just a power one — solder and sleeve it properly rather than relying on a twist.
- **Row 0 is the node's ground, row 8 is the cart's.** They are one net through L2, but keeping the node's return on its own strip means the cart's amps and the

modem's bursts are not sharing copper with the rail the ADC measures against — worth more than the one jumper it costs.

- **The right bank is free, but the regulator leans over it.** Nothing needs it electrically. If you later want slack on ground, one jumper from row 8 across the